

H2 Receptors: PPI's

1. "pour it all": parietal cells in the gastric mucosa are responsible for acid secretion
2. Battery powered puree pump: H+/K+ ATPase (proton pump) on the luminal membrane pumps H+ into the lumen
3. Banana into the pump, lemons out: the H+/K+ ATPase exchanges one K+ for one H+ at the luminal membrane
4. Sidewalk: lumen of the stomach
5. Three P batters: ATPase driven process
6. Bees swarming honey pot: histamine (released by the ECL cell) activates H2 receptors on the parietal cell
7. Honey pot with 2 "S" handles: H2 histamine receptors (coupled to Gs to increase cAMP) on the basolateral membrane
8. Honey pot kid knocking over acid pitcher: activation of H2 receptors upregulates the H+/K+ ATPase → increased acid secretion
9. Enter CarefuLly: enterochromaffin –like (ECL) cells
10. Bees released from ECL tree: the ECL cell releases histamine (activates the parietal cells)
11. Gas Powered blower releasing bees from ECL tree: gastrin (released by G cells) stimulates the ECL Cell to release histamine
12. Gas truck releasing gas tanks: G cells release gastrin (Which stimulates ECL and parietal cells)
13. Gas powered acid pump: gastrin (released by G cells) stimulates the parietal cell to secrete H+ (minor effect) gastrin's major effect is release of Histamine from ECL cells
14. Motorcycle attached to stand in M3: M3 acetylcholine receptors are located on the parietal cell
15. Vegas sticker: vagal stimulation stimulates the parietal by the vagus nerve (M3 receptor)
16. Vegas Sticker: vagal stimulation stimulates the G cell to release gastrin (VIA GRP)
17. Gate release pull: gastrin releasing peptide (GRP) from the vagus nerve activates G Cells
18. **2 bee swatters: H2 histamine receptor antagonists** (ranitidine, cimetidine) inhibit acid secretion by parietal cells
19.
20. Tie dye t shirt: "-tidine" suffix of H2 receptor antagonists (ranitidine, cimetidine, famotidine, nizatidine) @ antagonists reduce the acid secretion that is mediated by histamine, this comes from the enterocrhomaffin like cell, this cell is initially by gastrin by the G cell, which is brought on by vagal stimulation. So H2 blockers work with Histamine, gastrin, and Vagal stimulation. But Vagal stimulation will also stimulate the parietal cell directly to secrete acid. H2 blockers are used at night to prevent nocturnal secretion of acid that is largely dependent on histamine, but only a modest effect on meal stimulation because vagus nerve will stimulate Parietal cells directly
21. Gargling: H2 blockers (ranitidine, cimetidine) treat GERD (PPI's are first line)
22. Ulcerated sidewalk: H2 blockers (ranitidine, cimetidine) treat duodenal ulcers (PPI's are first line)
23. **Tie dye kid on the cement: cimetidine** (H2 blocker with antiandrogenic side effects)
24. Dented chrome bumper: Cimetidine inhibits cytochrome P-450
25. Pot lids on chest: cimetidine can cause gynecomastia when used long term or in high doses
26. Droopy honey wand: cimetidine can cause impotence
27. Milk shooting from nose: cimetidine can cause elevated serum prolactin levels
28. **Girl scout blocking puree pump: Proton Pump Inhibitors** (PPI's) irreversibly inhibit the H+/K+ ATPase (the final common pathway for H+ secretion)
29. PRIZE: "-prazole" suffix of PPI's (omeprazole, lansoprazole, rabeprazole)
30. Gargling: H2 blockerrs (ranitidine, cimetidine) treat GERD but PPI's are the first line
31. Ulcerated sidewalk: PPI's provide faster symptom relief for gastric and duodenal ulcers
32. Jumbo gas tank on mower with crab logo: gastrinoma causing hypersecretion of gastric acid (Zollinger-Ellison syndrome – treat with PPI's)
33.



34. Swarm of bees over lawn mower: extra gastrin will stimulate parietal cell directly with increasing Histamine, PPI's are the best medical treatment
35. Helicopter hat: PPI's treat H.Pylori infection, with clarithromycin or amoxicillin/metronidazole
36. Adverse effects
37. Chocolate fondue fountain: PPI's increase the risk for C.Diff infection
38. Dirty lung spots: PPI's increase the risk for respiratory infections (pneumonia)
39. Medals bound to wagon: PPI's decrease the absorption of Ca2+, Mg2+, and Fe2+ (requires acidic environment)
40. Fractured Axel: PPI's increase the risk of osteoporotic hip fractures (due to decreased Ca2+ absorption)
41. Porous wood: PPI's may worsen osteoporosis (due to decreased Ca2+ absorption)
42. Falling magnets on girl scout: PPI's can cause hypomagnesia
43. **Stop sign: Somatostatin** inhibits release of histamine by ECL cells
44. Stop sign: Somatostatin (SST) inhibits the release of gastrin by G cells (and SST receptor positive gastrinomas)
45. **Octagon shape of stop sign: octreotide** (a long acting SST analog) inhibits ECL production
46. Octagon: octreotide (a long acting SST analog) inhibits G cells (useful in the treatment of gastrinoma/Zollinger Ellison syndrome

